Morphologically-aware Tokenization

- In morphologically-rich languages, especially ones where the morphology is not concatenative, the use of morphological segmentation may be helpful
- For example, the Arabic-centric Fanar model uses a modification to BPE, motivating it with this example:

الرحمن ← ال + رح + من

In Hebrew transliteration: **אל-רחמן ← אל + רח + מן**

https://arxiv.org/pdf/2501.13944

Morphologically-aware Tokenization

Algorithm 1 Morphologically-aware BytePair Encoding (MorphBPE)

```
Initialize vocabulary with individual characters
Segment the training corpus using morphological segmentation
while number of merges < desired vocabulary size do
    Compute byte-pair frequencies
    Modified Step: Find the most frequent byte pair that does not cross morpheme boundaries
    Merge the most frequent byte pair into a new symbol
    Update the vocabulary with the merged symbol
end while
```

الرحمن ← ال + رحمن (אל + רחמן)

- This way, cases of letter sequences that are incomplete parts of morphemes will not be counted with their instances as complete morphemes

Morphologically-aware Tokenization

- Similar benefits from preceding the stastistical tokenization with morphological segmentation are shown for Hebrew (in this case WordPiece and not BPE)

Prefix	**Suffix**	**Form**	**English Translation**	**WordPiece Tokenization**	**Morphological Segmentation Tokenization**	**Prefix-Suffix Separation Tokenization**
-	-	שחרור	liberation	שחרור	שחרור	ש+חרור
ש	-	ששחרור	that lib.	ששחר ##ור	ש+שחרור	ש+שחרור
ו	ה	ושחרורה	and her lib.	ושחרור ##ה	ו+שחרור+ה	ו+ש+חרור+ה
ו+כ	נו	וכשחרורנו	and as our lib.	וכש ##חרור ##נו	ו+כ+שחרור+נו	ו+כ+ש+חרור+נ+ו
(1)	(2)	(3)	(4)	(5)	(6)	(7)

https://arxiv.org/pdf/2311.00658